Design a Lunar Rover Landing System

The situation. Your rover has made it almost all the way to the Moon. It will touch down vertically at 3.0 m/s, and its landing legs must bring it to rest without throwing it back into the air.

Your challenge

Design the energy-absorbing part of the landing gear. Begin with an elastic leg with no damping and explain why the rover rebounds. Then propose a system that dissipates the landing energy—for example a hydraulic dashpot, friction damper, crushable cartridge, or a justified alternative.

Your design must:

- use no more than 180 mm of working travel inside a 200 mm mechanism;
- keep the footpads on the surface after first contact;
- settle within 2.0 s; and
- survive the resulting axial, bending, and buckling loads with a safety factor of at least 2.

Choose a suitable structural material yourself. State its elastic modulus, yield strength, density, and the source of those properties.

Given information

Quantity	Value	Unit
Design touchdown mass supported by the legs, m	180	kg
Lunar gravity, g_{Moon}	1.62	m/s^2
Design vertical touchdown speed, v_0	3.0	m/s
Landing legs	4	—
Legs sharing the worst-case load	2	—
Structural leg length, L	0.70	m
Load eccentricity for bending, e	30	mm

Load sharing. The rover has four legs. A level touchdown on flat ground gives $F_{\text{leg}} = F_{\text{gear,total}}/4$. For the required worst-case structural check, assume two heavily loaded legs: $F_{\text{leg}} = F_{\text{gear,total}}/2$.

If you use a linear spring and dashpot, start from

$$m\ddot{x} + c\dot{x} + kx = mg, \qquad F_{\text{gear}} = kx + c\dot{x}, \qquad \zeta = \frac{c}{2\sqrt{mk}}.$$

